

## MSMP configuration in SAP GRC Access Control:

SAP GRC 12.0 Access Control standardizes SAP business workflow technology and supports more flexible and tailored access requests and approver views. It also helps to simplify the provisioning process.

This is where the SAP GRC 12.0 User Provisioning tool can help. The key engine that drives this process of the approval workflow is called **MSMP- Multi Step Multi Process**.

### Prerequisites:

The following configuration should have been completed as part of the initial post-installation steps:

1. GRC\_MSMP\_CONFIGURATION BC Set has been enabled
2. Perform Automatic Workflow Customizing
3. Perform Tasks Specific Customizing
4. Activate Event Linkage
5. Define number ranges for Access Requests
6. Connectors assigned to the PROV integration scenario

### MSMP Configuration:

Run T-Code SPRO and navigate to GRC---Access Control---Workflow for Access Control---Maintain MSMP Workflows.



These 7 steps allow you to customize and maintain the Multi-Stage Multi-Path (MSMP) process workflows for Access Control 12.0.

**NOTE: Before proceeding please activate GRC\_MSMP\_CONFIGURATION BC Set.**

#### 1. Process Global Settings:

The Process Global settings step uses the Initiator Rule for the Process Id (i.e. SAP\_GRAC\_ACCESS\_REQUEST). Each Process ID has one and only one active Initiator Rule.

Display / Change

1 2 3 4 5 6 7

Process Global Settings Maintain Rules Maintain Agents Variables & Templates Maintain Paths Maint Route Mapping Generate Versions

Select workflow process

| Process ID                    | Process Description                                | Escalation | Rule ID                     | Rule Description                                     |
|-------------------------------|----------------------------------------------------|------------|-----------------------------|------------------------------------------------------|
| SAP_GRAC_ACCESS_REQUEST       | Access Request Approval Workflow                   |            | GRAC_AR_INITIATOR           | Default Initiator rule ( Process Type : SAP_GRAC_AR) |
| SAP_GRAC_ACCESS_REQUEST_HR    | Access Request Approval for HR OM Objects Workflow |            | GRAC_AR_INITIATOR           | Default Initiator rule ( Process Type : SAP_GRAC_AR) |
| SAP_GRAC_CONTROL_ASGN         | Control Assignment Approval Workflow               |            | GRAC_CTRLASGN_INITIATOR     | Default Initiator rule                               |
| SAP_GRAC_CONTROL_MAINT        | Mitigation Control Maintenance Workflow            |            | GRAC_MITCTRLMAINT_INITIATOR | Default Initiator rule                               |
| SAP_GRAC_FFID_REVIEW          | Firefighter ID Review Workflow                     |            | GRAC_FFREVIEW_INITIATOR     | FF Review : Initiator Rule                           |
| SAP_GRAC_FIREFIGHT_LOG_REPORT | Fire Fighter Log Report Review Workflow            |            | GRAC_FFLOGREPORT_INITIATOR  | Default Initiator rule                               |
| SAP_GRAC_FUNC_APPR            | Function Approval Workflow                         |            | GRAC_FUNCAPPR_INITIATOR     | Default Initiator rule                               |
| SAP_GRAC_RISK_APPR            | Risk Approval Workflow                             |            | GRAC_RISKAPPR_INITIATOR     | Default Initiator rule                               |
| SAP_GRAC_ROLE_APPR            | Role Approval Workflow                             |            | GRAC_ROLEAPPR_INITIATOR     | Default Initiator rule                               |
| SAP_GRAC_SOD_RISK_REVIEW      | SOD Risk Review Workflow                           |            | GRAC_RISKREVIEW_INITIATOR   | Default Initiator rule                               |

Process Global Settings

Notification Settings

| Notification Event | Template ID | Recipient ID |
|--------------------|-------------|--------------|
|--------------------|-------------|--------------|

## Initiator Rule

- Initiate the request
- There will be unique conditions for the Initiator rule
- There can be only one initiator rule for each Process ID in MSMP configuration
- In case there is a custom requirement a BRF Plus rule needs to be created and these additional setting has to be done
- The rule and its available results must be declared
- Routings must be maintained

## Routing Rules

- Similar to Initiator rules, except that they are activated during the workflow process
- Additional configuration is required to finalize the use of Routing Rules in the MSMP workflow
- The rule and its available results must be declared & Routings must be maintained

## 2. Maintain Rules:

Maintain Rules includes a list of all available rules to be used when configuring a workflow. If a new rule is created it must be added to this list. This is also where the default initiator is configured.

## MSMP Workflow Configuration: Step 2 of 7 (Maintain Rules)

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| 1                             | 2              | 3                                                    | 4                     | 5                          | 6                   | 7                 |
|-------------------------------|----------------|------------------------------------------------------|-----------------------|----------------------------|---------------------|-------------------|
| Process Global Settings       | Maintain Rules | Maintain Agents                                      | Variables & Templates | Maintain Paths             | Maint Route Mapping | Generate Versions |
| GRAC_MSMP_SPM_OWNER_AGENT     |                | SPM owner agent ( Process Type : SAP_GRAC_AR)        |                       | Function Module Based Rule |                     | Agents Rule       |
| GRAC_MSMP_USER_AGENT          |                | User agent ( Process Type : SAP_GRAC_AR)             |                       | Function Module Based Rule |                     | Agents Rule       |
| GRFNMW_INST_CURRENT_APPROVERS |                | Current approvers ( Process Type : SAP_GRAC_AR)      |                       | Function Module Based Rule |                     | Agents Rule       |
| GRAC_AR_INITIATOR             |                | Default Initiator rule ( Process Type : SAP_GRAC_AR) |                       | Function Module Based Rule |                     | Initiator Rule    |
| GRAC_DETOUR_SODVIOL_NO_ROLOWN |                | SOD violation and no roleowner chained routing rule  |                       | Function Module Based Rule |                     | Routing Rule      |

### Rule Results

Add Modify Delete

| *Rule Result Value  | Trigger Value Descr       |
|---------------------|---------------------------|
| GRAC_DEFAULT_RESULT | Default rule result value |

Cancel

### Global Rules

|                   |                        |                                                          |
|-------------------|------------------------|----------------------------------------------------------|
| Process Initiator | GRAC_AR_INITIATOR      | Default Initiator rule ( Process Type : SAP_GRAC_AR)     |
| Notification Rule | GRAC_NOTIF_VAR_RULE_AR | Notification variables rule( Process Type : SAP_GRAC_AR) |

It is required to maintain a list of all possible results returned by an initiator/routing rule by using the Results button. These values will be mapped to a path in step 6.

## 3. Maintain Agents:

### Agent Rules:

The persons that will either be notified or approved during the workflow. Conditions analyzed for an agent rule do not have to be unique because the rule can return multiple results if the business process requires such. The rule must be declared as an agent, including its purpose. The Agent can be defined as GRC API rule, PFCG User Group, PFCG Role, Directly Mapped users.

## MSMP Workflow Configuration: Step 3 of 7 (Maintain Agents)

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| 1                       | 2              | 3               | 4                     | 5              | 6                   | 7                 |
|-------------------------|----------------|-----------------|-----------------------|----------------|---------------------|-------------------|
| Process Global Settings | Maintain Rules | Maintain Agents | Variables & Templates | Maintain Paths | Maint Route Mapping | Generate Versions |

Process : Access Request Approval Workflow

### Agents

Add Modify Delete

View Filter Fields Export

| Agent ID                      | Agent Name                                           | Agent Type                 | Agent Purpose  |
|-------------------------------|------------------------------------------------------|----------------------------|----------------|
| GRAC_AR_INITIATOR             | Default Initiator rule ( Process Type : SAP_GRAC_AR) | Function Module Based Rule | Initiator Rule |
| GRAC_MSMP_SPM_OWNER_AGENT     | SPM owner agent ( Process Type : SAP_GRAC_AR)        | Function Module Based Rule | Agents Rule    |
| GRAC_MSMP_USER_AGENT          | User agent ( Process Type : SAP_GRAC_AR)             | Function Module Based Rule | Agents Rule    |
| GRFNMW_INST_CURRENT_APPROVERS | Current approvers ( Process Type : SAP_GRAC_AR)      | Function Module Based Rule | Agents Rule    |
| GRAC_AR_INITIATOR             | Default Initiator rule ( Process Type : SAP_GRAC_AR) | Function Module Based Rule | Initiator Rule |
| GRAC_DETOUR_SODVIOL_NO_ROLOWN | SOD violation and no roleowner chained routing rule  | Function Module Based Rule | Routing Rule   |

\* Agent ID

\* Agent Name

\* Agent Purpose

\* Agent Type

Save Cancel

Directly Mapped Users

PFCG Roles

PFCG User Groups

GRC API Rules

## 4. Variables and Templates:

### Notification and Variables Rules:

Identifies the possible variables that can be used in the request notification template. Additional configuration is required to finalize the use of the Notification and Variables rule. The rule and variables must be declared.

#### MSMP Workflow Configuration: Step 4 of 7 (Variables & Templates)

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|           |                       |                     |                |                             |       |
|-----------|-----------------------|---------------------|----------------|-----------------------------|-------|
| AddDelete | Template ID           | Message Class       | Message number | Docu Object                 | Owner |
|           | GRAC_AR_APPROVED      | 0MSMP_AR_APPROVED   | 000            | GRAC_MSMP_AR_APPROVED       | SAP   |
|           | GRAC_AR_APP_BY_EMAIL  | 0MSMP_AR_APP_REJ    | 000            | GRAC_MSMP_AR_NEWWORKITM_APP | SAP   |
|           | GRAC_AR_CLOSE         | 0AC_AR_CLOSE        | 000            | GRAC_AR_CLOSE               | SAP   |
|           | GRAC_AR_ESCALATION    | 0MSMP_AR_ESCALATION | 000            | GRAC_MSMP_AR_ESCALATION     | SAP   |
|           | GRAC_AR_FORWARD       | 0MSMP_AR_FORWARD    | 000            | GRAC_MSMP_AR_FORWARD        | SAP   |
|           | GRAC_AR_NEW_WORK_ITEM | 0MSMP_AR_NEWWORKITM | 000            | GRAC_MSMP_AR_NEWWORKITM     | SAP   |
|           | GRAC_AR_REJECTED      | 0MSMP_AR_REJECTED   | 000            | GRAC_MSMP_AR_REJECTED       | SAP   |
|           | GRAC_AR_SUBMIT        | 0AC_AR_SUBMIT       | 000            | GRAC_AR_SUBMIT              | SAP   |
|           | GRAC_EMAILRMDR_CUP    | 0MSMP_EMAILRMDR_CUP | 000            | GRFNMW_EMAILRMDR_CUP        | SAP   |

#### Notification Variables

|                 |                |                      |
|-----------------|----------------|----------------------|
| AddModifyDelete | Temp. Variable | Variable Description |
|                 | APPROVED_BY    | Request Approver     |
|                 | FIRST_NAME     | Requester First Name |
|                 | LAST_NAME      | Requester Last Name  |

## 5. Maintain Paths:

In this step, we can specify the approval agent. We can decide if the approval path will be one stage or two stages. Here we can specify the path ID and agent ID. In this setting, we look for the GRAC\_DEFAULT\_PATH and identify the Stage in the Path. This shows that there are two stages which are manager and Security. Now we need to understand how the approvers are assigned to the stages.

#### MSMP Workflow Configuration: Step 5 of 7 (Maintain Paths)

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|                 |                   |                        |
|-----------------|-------------------|------------------------|
| AddModifyDelete | GRAC_DEFAULT_PATH | Default Path           |
|                 | Z_CHANGE_ACCESS1  | Change Existing Access |
|                 | Z_CREATE_ACCESS1  | Create New Access      |
|                 | Z_DELETE_ACCESS1  | Delete Access          |
|                 | Z_LOCK_ACCESS1    | Lock Access            |
|                 | Z_NOSTAGE_PATH    | Nostage Path           |
|                 | Z_SUPER_USER1     | super user             |
|                 | Z_UNLOCK_ACCESS1  | Unlock Access          |

Display Task SettingsNotification SettingsExpand Stage Path

Stage Seq No

Stage Config ID

Stage Des

Agent ID

Routing

Rule Type

Rule ID

Routing

Escalati

Escalati

Approva

Escalation

|     |                    |         |                |    |  |  |  |              |  |  |   |
|-----|--------------------|---------|----------------|----|--|--|--|--------------|--|--|---|
| 020 | GRAC_DEFAULT_STAGE | default | Z_ARM_APPROVER | No |  |  |  | Use Defaults |  |  | 0 |
|-----|--------------------|---------|----------------|----|--|--|--|--------------|--|--|---|

MSMP Workflow Configura... ×

### MSMP Workflow Configuration: Step 5 of 7 (Maintain Paths)

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| Add       | Modify          | Delete         | Modify Task Settings | Notification Settings | Clean-up Stage Def. |           |         |               |              |               |              |  |
|-----------|-----------------|----------------|----------------------|-----------------------|---------------------|-----------|---------|---------------|--------------|---------------|--------------|--|
| Stage Seq | Stage Config ID | Stage Descr    | Agent ID             | Approval Type         | Routing Ena         | Rule Type | Rule ID | Routing Level | Escalation T | Escalation Ti | Escalation A |  |
| 001       | GRAC_MAN...     | Manager st...  |                      |                       | No                  |           |         |               |              |               | 0            |  |
| 002       | GRAC_ROLE...    | Role owner...  |                      |                       | No                  |           |         |               |              |               | 0            |  |
| 003       | GRAC_SECU...    | Security stage |                      |                       | No                  |           |         |               |              |               | 0            |  |

Stage Seq No: 001

Stage Config ID: GRAC\_MANAGER

Stage Description: Manager stage

Hide Details

Agent ID: GRAC\_MANAGER Manager

Approval Type: Any One Approver

Routing Enabled: ☒

Rule ID: GRAC\_DETOUR\_SODVIOL\_NO\_R... SOD violation and no roleowner chained routing rule ( Process Type: SAP\_GRAC\_AR)

Routing Level: Stage Level

Escalation Type: Use Defaults

Save Cancel

Use Defaults

Skip To Next Stage

## 6. Maintain Route Mapping:

From Step 2 we added the Rule Result value i.e. GRAC\_DEFAULT\_RESULT and Path Id is GRAC\_DEFAULT\_PATH to which it is mapped for.

Always the Global Initiator must be used, if multiple paths are required the Global The initiator must return different result values

### MSMP Workflow Configuration: Step 6 of 7 (Maint Route Mapping)

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Process: Access Request Approval Workflow

#### Maintain Route Mapping

Add Modify Delete

View: [Standard View] Export

| Rule ID           | Rule Result Value   | From Path ID | From Stage | Path ID           |
|-------------------|---------------------|--------------|------------|-------------------|
| GRAC_AR_INITIATOR | GRAC_DEFAULT_RESULT |              |            | GRAC_DEFAULT_PATH |

7. Generate Versions:

In the last step, all changes will be saved and activated.

MSMP Workflow Configuration: Step 7 of 7 (Generate Versions)

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Maintain Paths

Maint Route Mapping

Generate Versions

Process: Access Request Approval Workflow

Save & Activate

Save

Save / Simulate

Activate

View: [Standard View]

Export

T...

Message text